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# DATA SCIENCE PROJECT: MARKET BILL ANAL
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# STEP 1: IMPORT LIBRARIES

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import trai
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import Standa
from sklearn.linear_model import LinearR
from sklearn.metrics import mean_absolut


# STEP 2: CREATE MARKET BILL DATASET

data = pd.DataFrame({
    "Product": [
        "Rice", "Sugar", "Oil", "Milk",
        "Rice", "Sugar", "Oil", "Milk",
        "Rice", "Sugar", "Oil", "Milk",
        "Rice", "Sugar", "Oil", "Milk",
    ],
    "Quantity": [
        2, 3, 1, 4, 2,
        3, 2, 2, 5, 3,
        4, 1, 3, 2, 4,
        5, 4, 2, 6, 3
    ],
    "Price": [
        60, 45, 150, 30, 40,
        60, 45, 150, 30, 40,
        60, 45, 150, 30, 40,
        60, 45, 150, 30, 40
    ]
})

print("========== ORIGINAL DATASET ====
print(data)


# STEP 3: DATA MANIPULATION

# Calculate product amount
data["Amount"] = data["Quantity"] * dat
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data["Amount"] = data["Quantity"] * data[

# Remove duplicate rows
data = data.drop_duplicates().copy()

print("\n========== DATA INFORMATION ===
data.info()

print("\n========== STATISTICAL SUMMARY
print(data.describe())

# Product-wise sales summary
summary = data.groupby("Product").agg(
    Total_Quantity=("Quantity", "sum"),
    Price=("Price", "first"),
    Total_Amount=("Amount", "sum")
)

print("\n========== PRODUCT-WISE SALES =
print(summary)

print("\nTotal Market Bill: Rs.", data['


# STEP 4: DATA VISUALIZATION

# Bar chart
summary["Total_Amount"].plot(
    kind="bar",
    figsize=(8, 4)
)

plt.title("Product-wise Market Sales")
plt.xlabel("Product")
plt.ylabel("Total Amount (Rs.)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Scatter plot
sns.scatterplot(
    data=data,
    x="Quantity",
    y="Amount",
    hue="Product"
)

plt.title("Quantity vs Product Amount")
plt.tight_layout()
plt.show()

# Correlation heatmap
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plt.figure(figsize=(6, 4))

sns.heatmap(
    data[["Quantity", "Price", "Amount"]
    annot=True,
    cmap="coolwarm"
)

plt.title("Market Bill Correlation")
plt.tight_layout()
plt.show()


# STEP 5: DATA PREPROCESSING

# Input features and target
X = data[["Quantity", "Price"]]
y = data["Amount"]

print("\n========== MISSING VALUES =====
print(data.isnull().sum())


# STEP 6: SPLIT THE DATASET

X_train, X_test, y_train, y_test = train
    X,
    y,
    test_size=0.25,
    random_state=42
)

print("\nTraining records:", len(X_train
print("Testing records:", len(X_test))


# STEP 7: HANDLE MISSING VALUES

imputer = SimpleImputer(strategy="median

X_train = imputer.fit_transform(X_train)
X_test = imputer.transform(X_test)


# STEP 8: FEATURE SCALING

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_
X_test_scaled = scaler.transform(X_test)
```

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# STEP 9: MACHINE LEARNING MODEL

# Use Linear Regression
model = LinearRegression()

# Train the model
model.fit(X_train_scaled, y_train)

print("\nLinear Regression model trained


# STEP 10: MODEL EVALUATION

# Predict test data
y_pred = model.predict(X_test_scaled)

mae = mean_absolute_error(y_test, y_pred
rmse = np.sqrt(mean_squared_error(y_test
r2 = r2_score(y_test, y_pred)

print("\n========== MODEL EVALUATION ===
print("Actual Amounts:", list(y_test))
print("Predicted Amounts:", np.round(y_p
print("Mean Absolute Error:", round(mae,
print("Root Mean Squared Error:", round(
print("R2 Score:", round(r2, 3))


# STEP 11: PREDICT A NEW MARKET BILL

new_purchase = pd.DataFrame({
    "Quantity": [5],
    "Price": [60]
})

# Apply the same preprocessing
new_purchase_imputed = imputer.transform
new_purchase_scaled = scaler.transform(n

# Predict using the same trained model
predicted_amount = model.predict(new_pur

print("\n========== NEW MARKET BILL ====
print("Product: Rice")
print("Quantity: 5")
print("Price per unit: Rs. 60")
print("Predicted Amount: Rs.", round(pre
print("Actual Calculated Amount: Rs.", 5


# STEP 12: FINAL SUMMARY
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```
print("\n========== PROJECT SUMMARY ====
print("Total Market Sales: Rs.", data["A
print("Number of Products:", data["Produ
print("Machine Learning Model: Linear Re
print("Data Science Project Completed Su
```

	Product	Quantity	Price
0	Rice	2	60
1	Sugar	3	45
2	Oil	1	150
3	Milk	4	30
4	Bread	2	40
5	Rice	3	60
6	Sugar	2	45
7	Oil	2	150
8	Milk	5	30
9	Bread	3	40
10	Rice	4	60
11	Sugar	1	45
12	Oil	3	150
13	Milk	2	30
14	Bread	4	40
15	Rice	5	60
16	Sugar	4	45
17	Oil	2	150
18	Milk	6	30
19	Bread	3	40

========== DATA INFORMATION ==========

```
<class 'pandas.core.frame.DataFrame'>
Index: 18 entries, 0 to 18
Data columns (total 4 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Product   18 non-null     object
 1   Quantity  18 non-null     int64
 2   Price     18 non-null     int64
 3   Amount    18 non-null     int64
dtypes: int64(3), object(1)
memory usage: 720.0+ bytes
```

========== STATISTICAL SUMMARY ==========

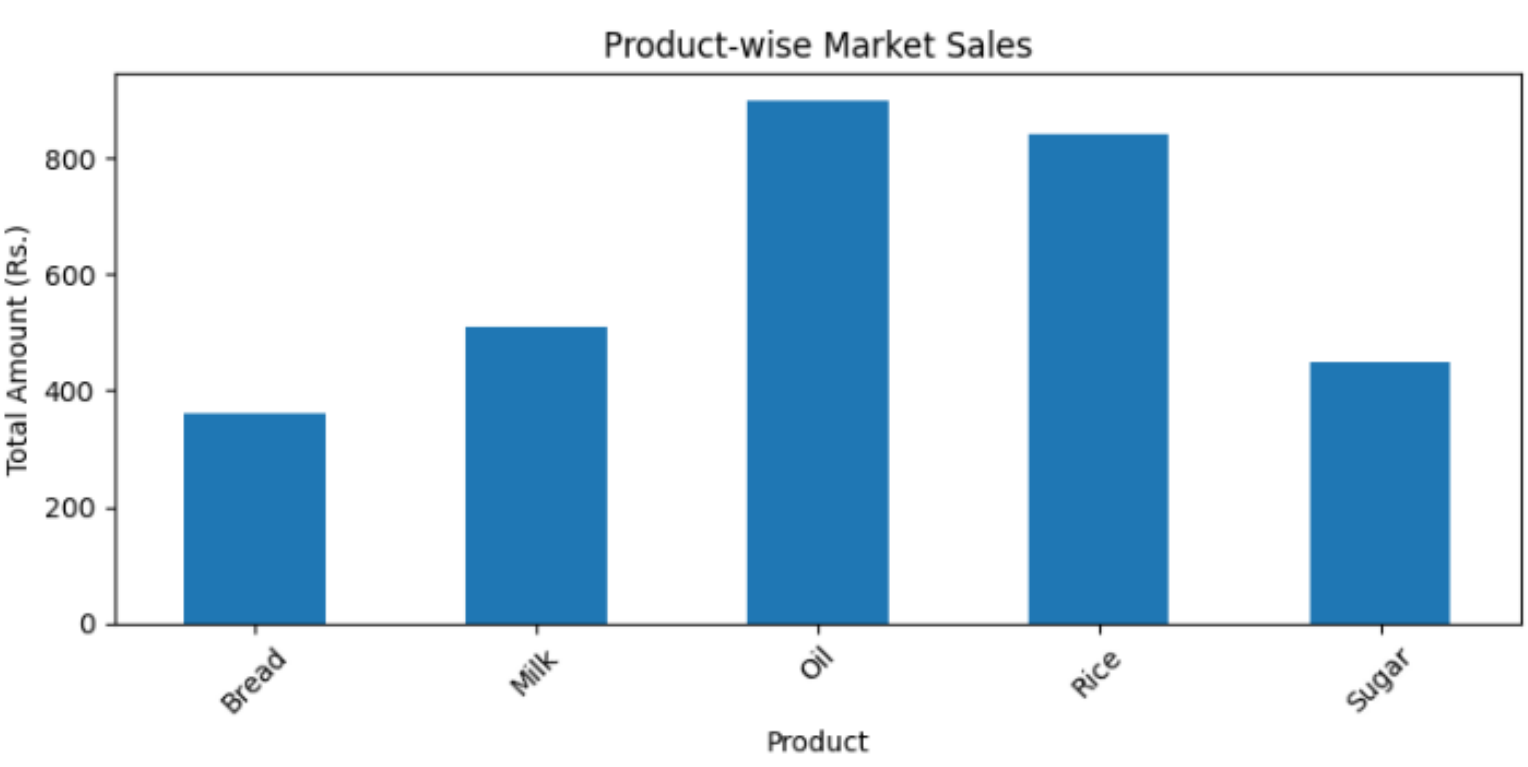
	Quantity	Price	Amount
count	18.000000	18.000000	18.000000
mean	3.111111	61.666667	170.000000
std	1.409584	41.973381	99.779168
min	1.000000	30.000000	45.000000
25%	2.000000	40.000000	120.000000
50%	3.000000	45.000000	150.000000
75%	4.000000	60.000000	180.000000
max	6.000000	150.0	

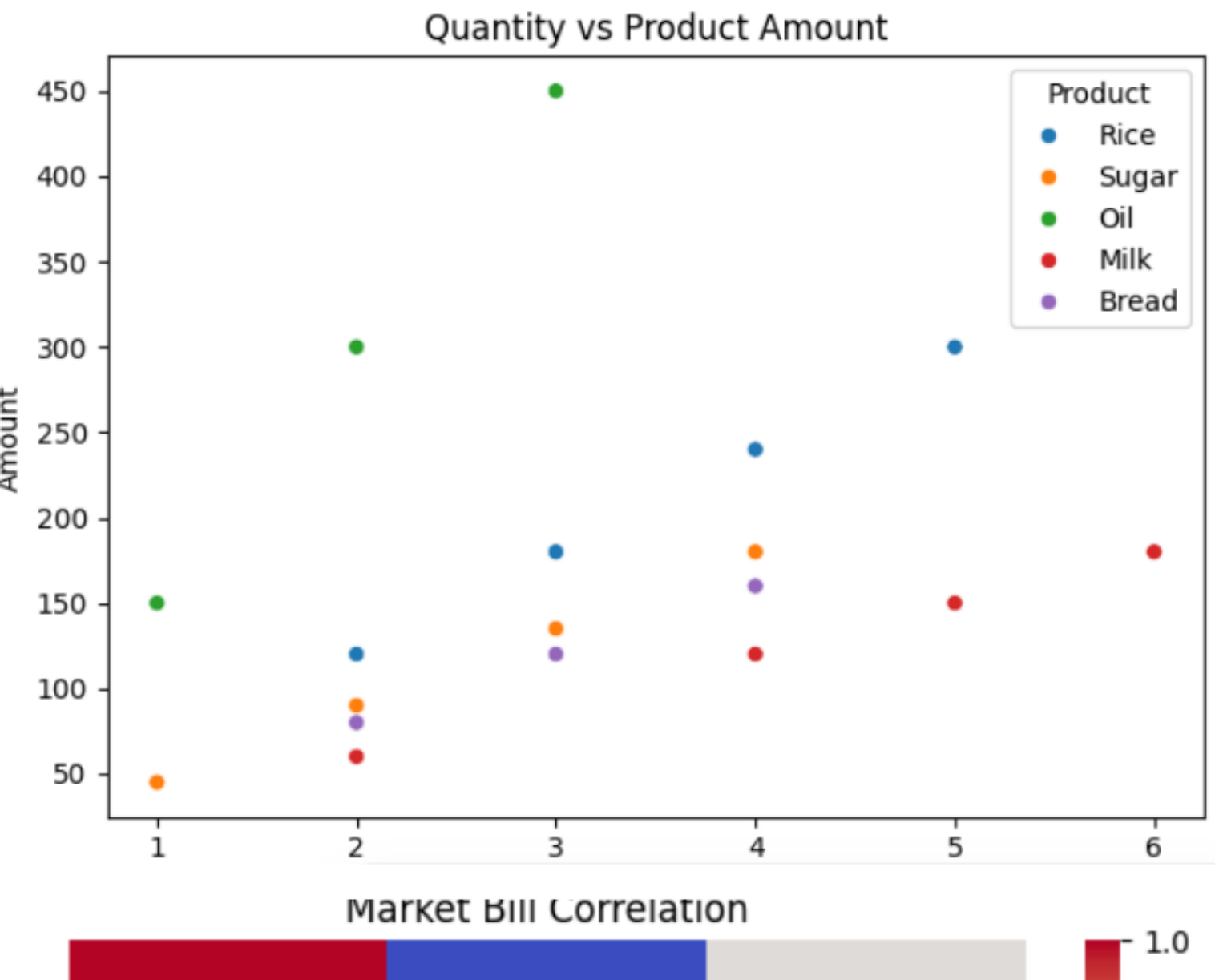
========== PRODUCT-WISE SALES ==========

Product	Total_Quantity	Price	Total_Amount
Bread	9	40	360
Milk	17	30	510
Oil	6	150	900
Rice	14	60	840
Sugar	10	45	450

Total Market Bill: Rs. 3060

Total Market Bill: Rs. 3060







========== MISSING VALUES ==========

```
Product     0
Quantity    0
Price       0
Amount      0
dtype: int64

Training records: 13
Testing records: 5
```

Linear Regression mode